

NYC Bus Route Optimisation

COS 805 Presentation



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Motivation / Problem Statement

Problem Statement

New York's bus network carries around **2.75 million trips a day**, yet over **90% of the streets with bus routes have no dedicated bus lanes**, and buses crawl through traffic at an average of about 8 mph.

Dedicated lanes are expensive and can't be built everywhere at once, so the city needs a **clear, data-driven way to decide where they would help the most people**.



Central question:

Which areas of New York City experience the most persistent and widespread congestion, using **yellow taxi trip speeds** as a proxy, and are therefore the strongest candidates for dedicated bus lanes?



Approach

- Millions of yellow taxi trip records are processed with PySpark
- The data is cleaned and filtered to valid trips, and create new features such as trip duration and average speed.
- The cleaned data then goes through a series of analyses to which areas stand to benefit most from dedicated bus lanes and more frequent buses.



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Data Cleaning

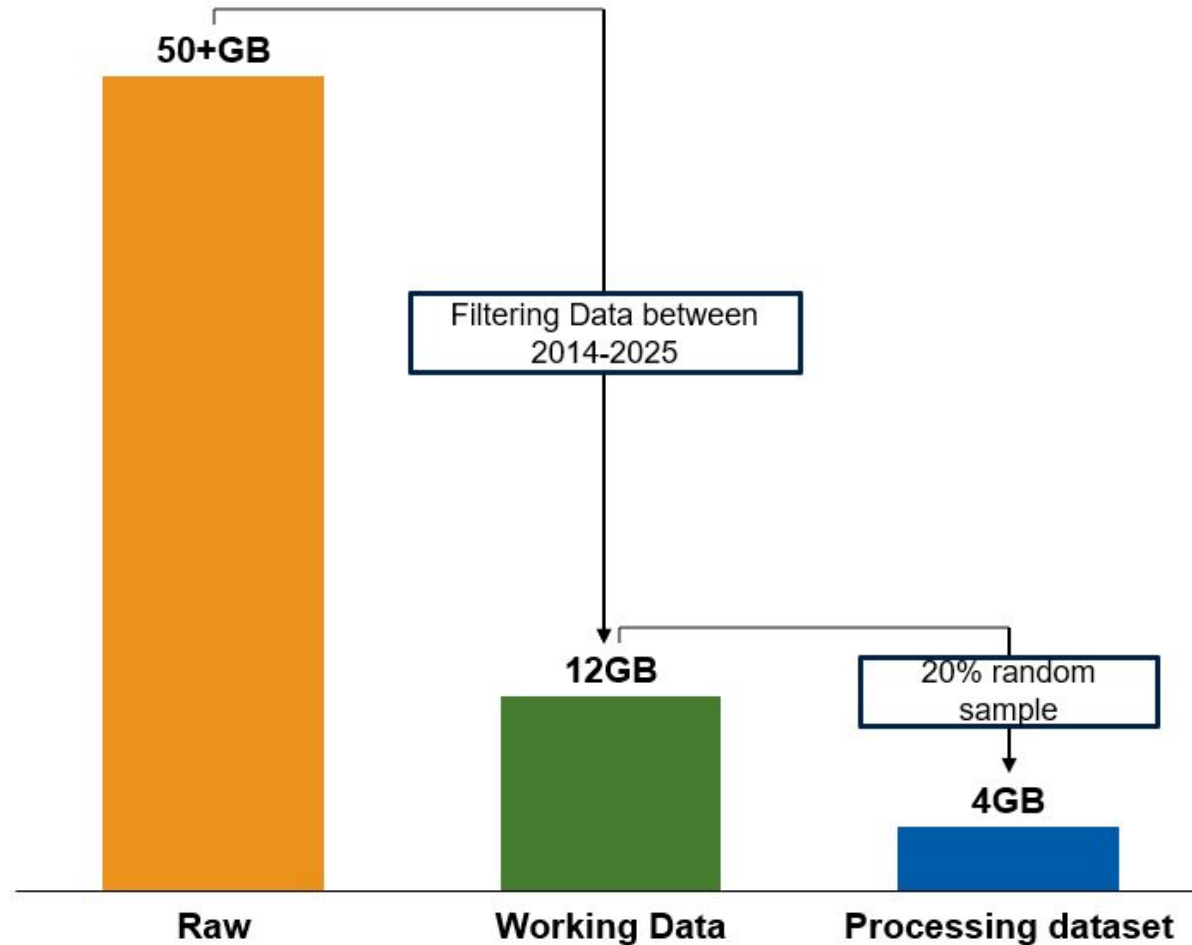


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Data Cleaning – Dataset Reduction



- The 50GB raw dataset was reduced down to 4GB by filtering for the selected period and taking a 20% random sample.
- **The dataset was cleaned in four steps:**
 - Standardised data types
 - Removed irrelevant columns
 - Removed unrealistic values and outliers such as negative distances.
 - Removed duplicates.



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Exploratory Data Analysis



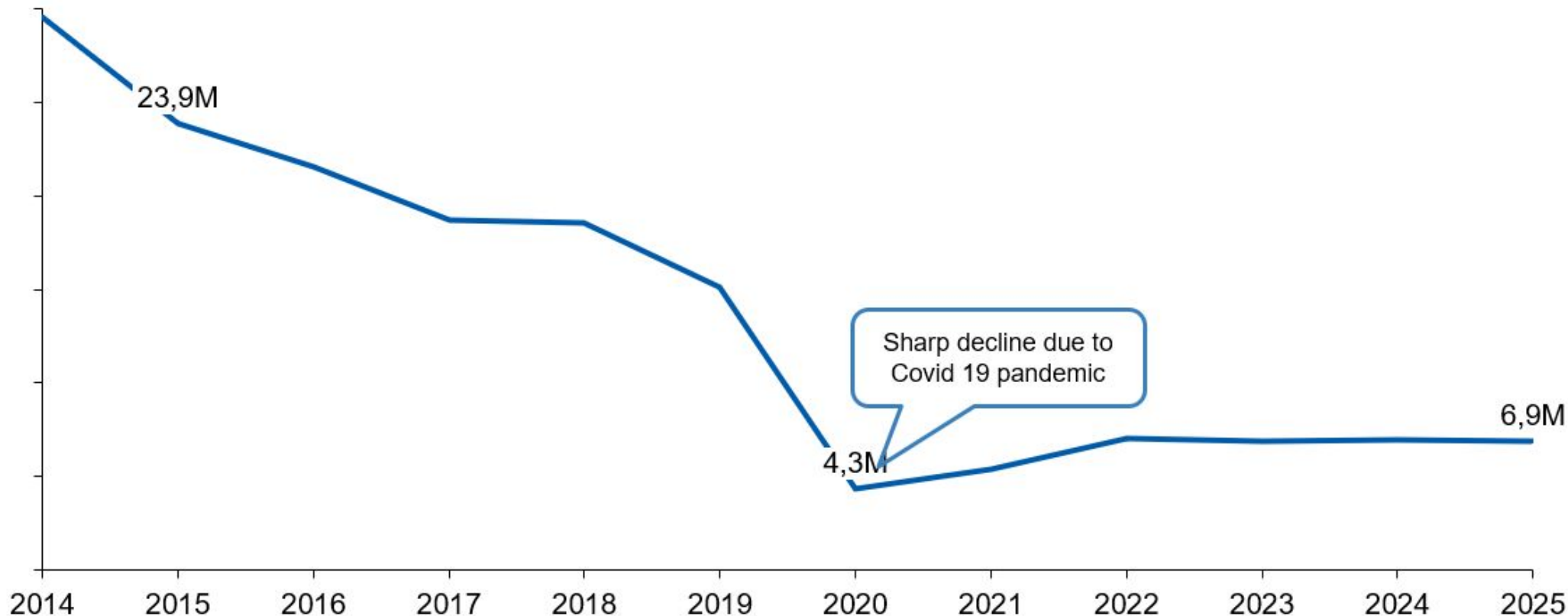
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Exploratory Data Analysis

Yellow cab trip volumes **gradually decreased between 2014 and 2019**, which could be attributed to untracked vendors or the reduced reliance on e-hailing services such as Uber. However, the **70%+ drop from 2019 to 2020 can be directly attributed to the COVID-19 pandemic** and the resulting decline in travel demand.

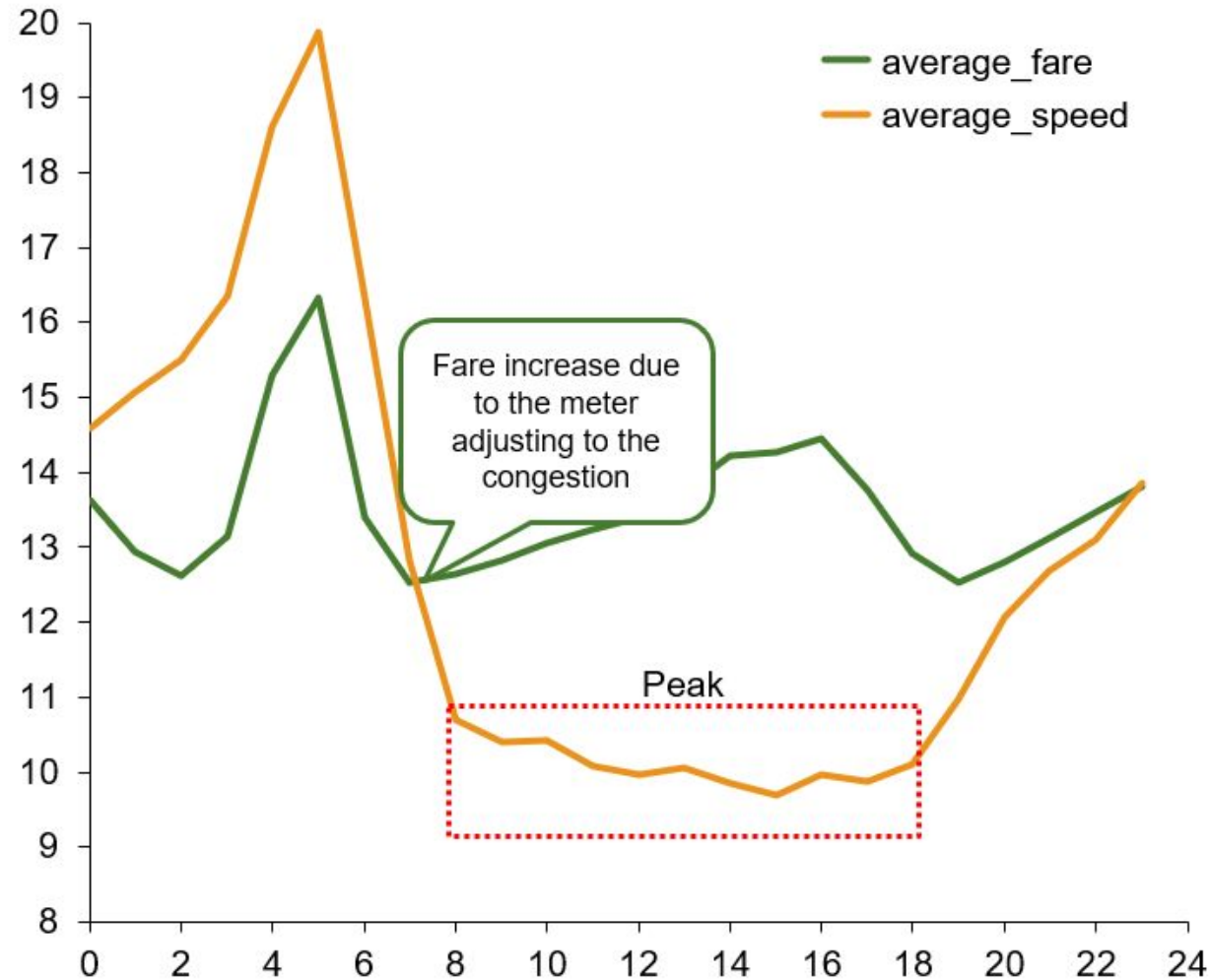


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Exploratory Data Analysis



- As roads get busier from 6-7am and speeds drop, **average fare climbs further still**. This is due to how the taxi meter is structured: fares are normally based on a flat per-mile rate, but below a certain speed, an additional **time-based fee is applied** to account for the slower progress
- Congestion is at its worst in the city between 8am and 6pm



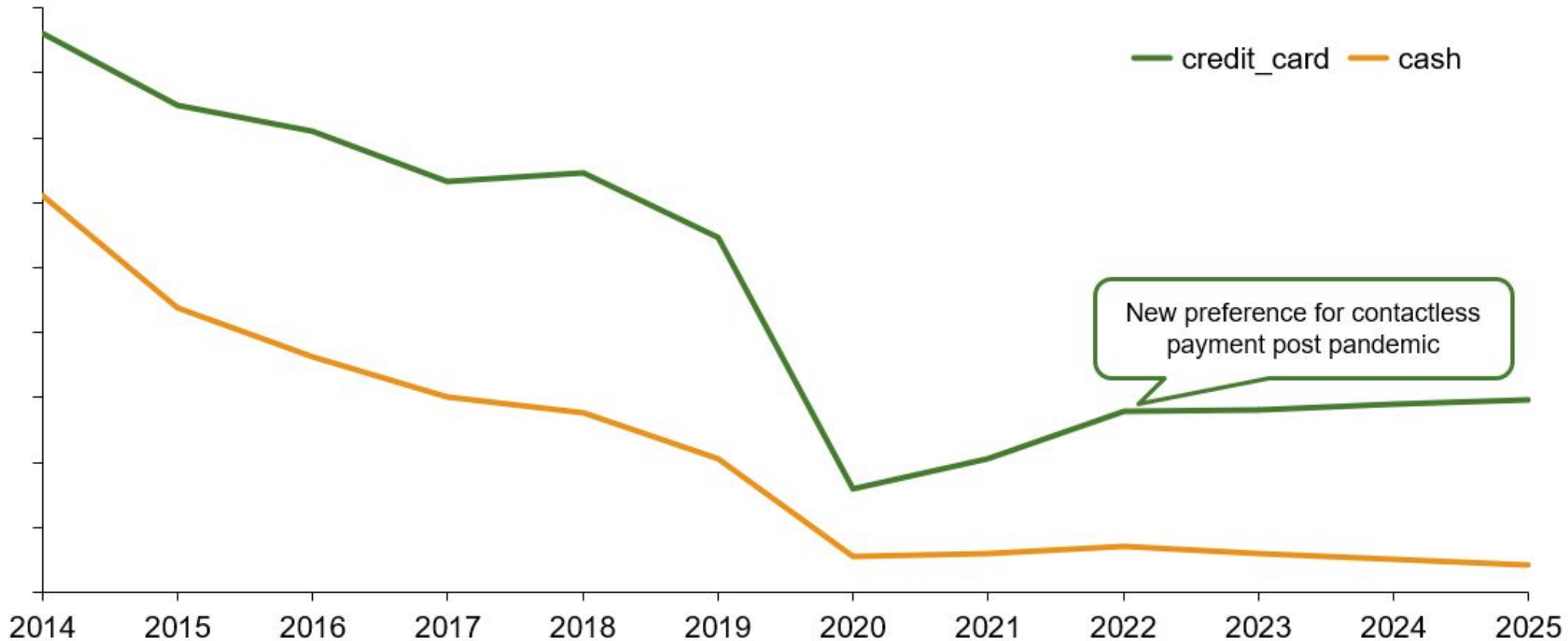
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Exploratory Data Analysis

Both cash and credit card payments saw a **decline between 2014-2020**, however, following **COVID-19**, **credit card usage gradually increased** while cash continued to decline, likely due to the **preference for contactless payments**.



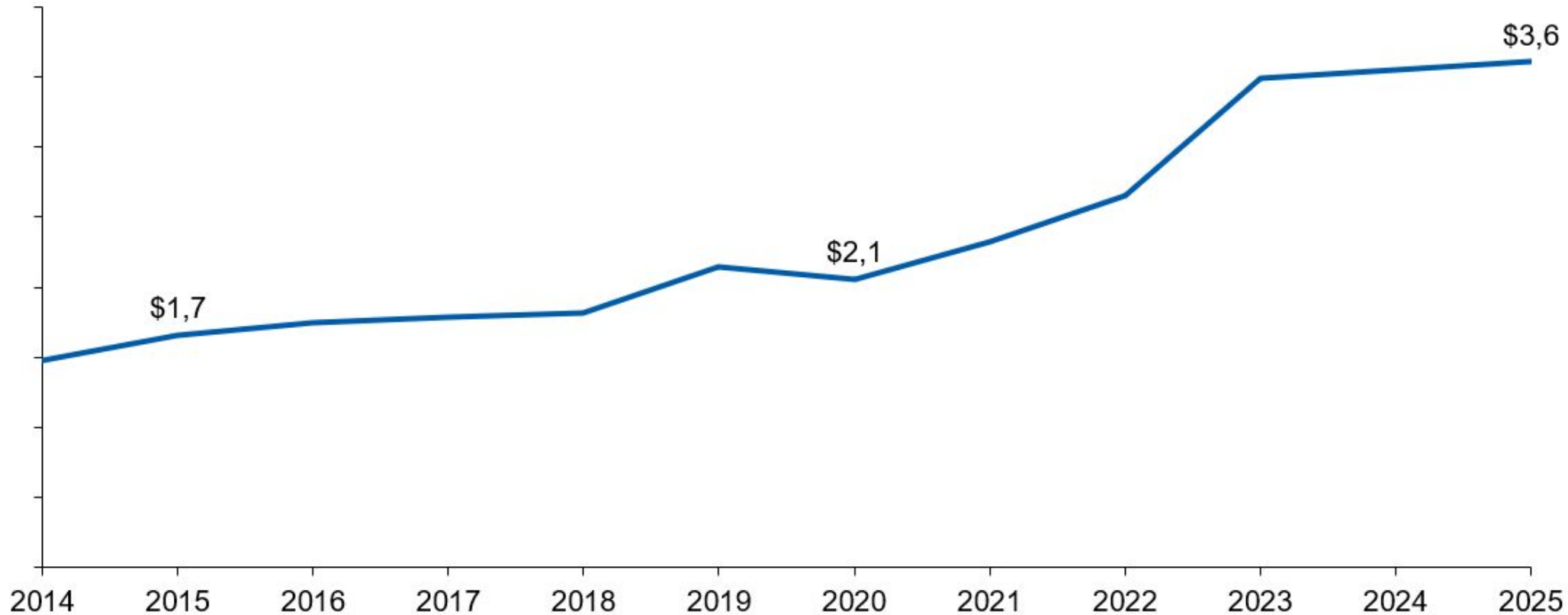
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Exploratory Data Analysis

The average tip per ride generally increases over time, possibly due to considerations such as inflation.



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Project Part 2: Bus Route Optimisation



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Project Part 2: Bus Route Optimisation

OBJECTIVES:

Analysis was completed using the MapReduce style implementation in PySpark to provide the following recommendations to the City Planning Office of New York City:

- Priority list of areas for implementation of additional bus routes or dedicated bus lanes.
- Economic recommendations on bus fares.
- Bus schedule recommendations.
- Recommendations on payment systems to use

These were done through studying the following two elements:

- Analysis of trends per area (i.e., trends in all of the New York City boroughs)
- Analysis of trends over time



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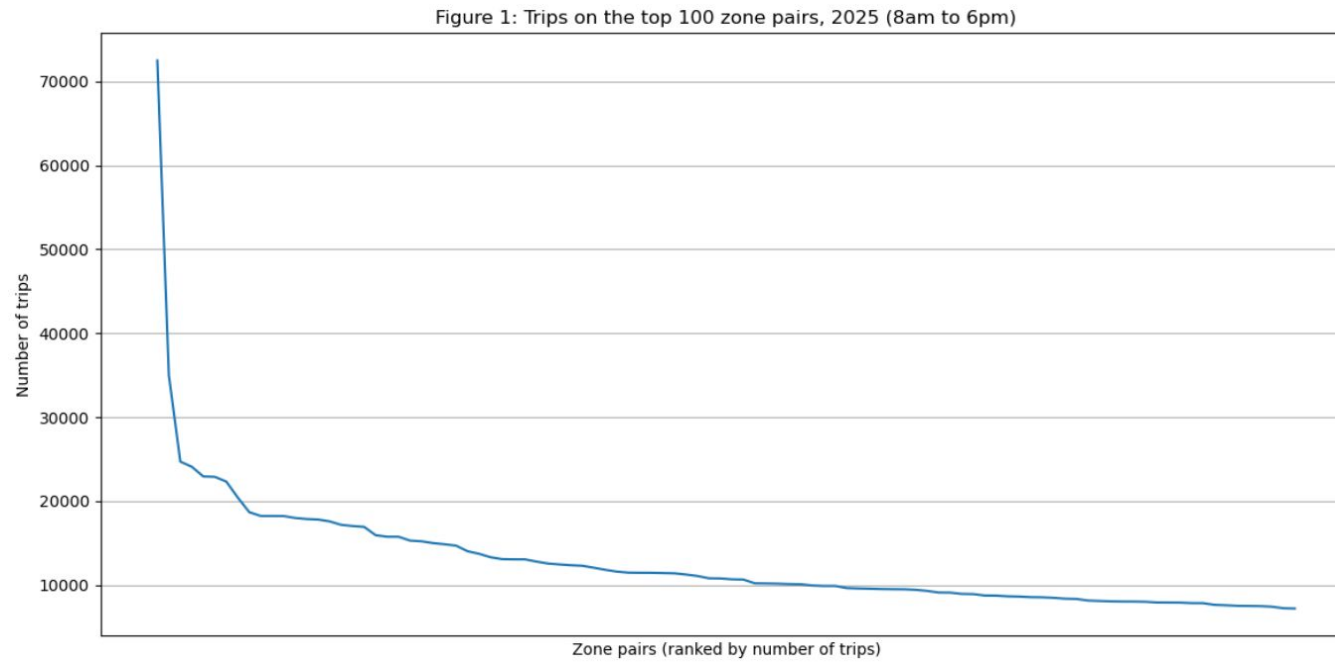
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RESULTS:

- Identification of routes for implementation of bus infrastructure (dedicated bus lanes, more frequent bus routes etc).
- Routes greater than 1 mile analysed by zone pair
- Small trips less than 1 mile



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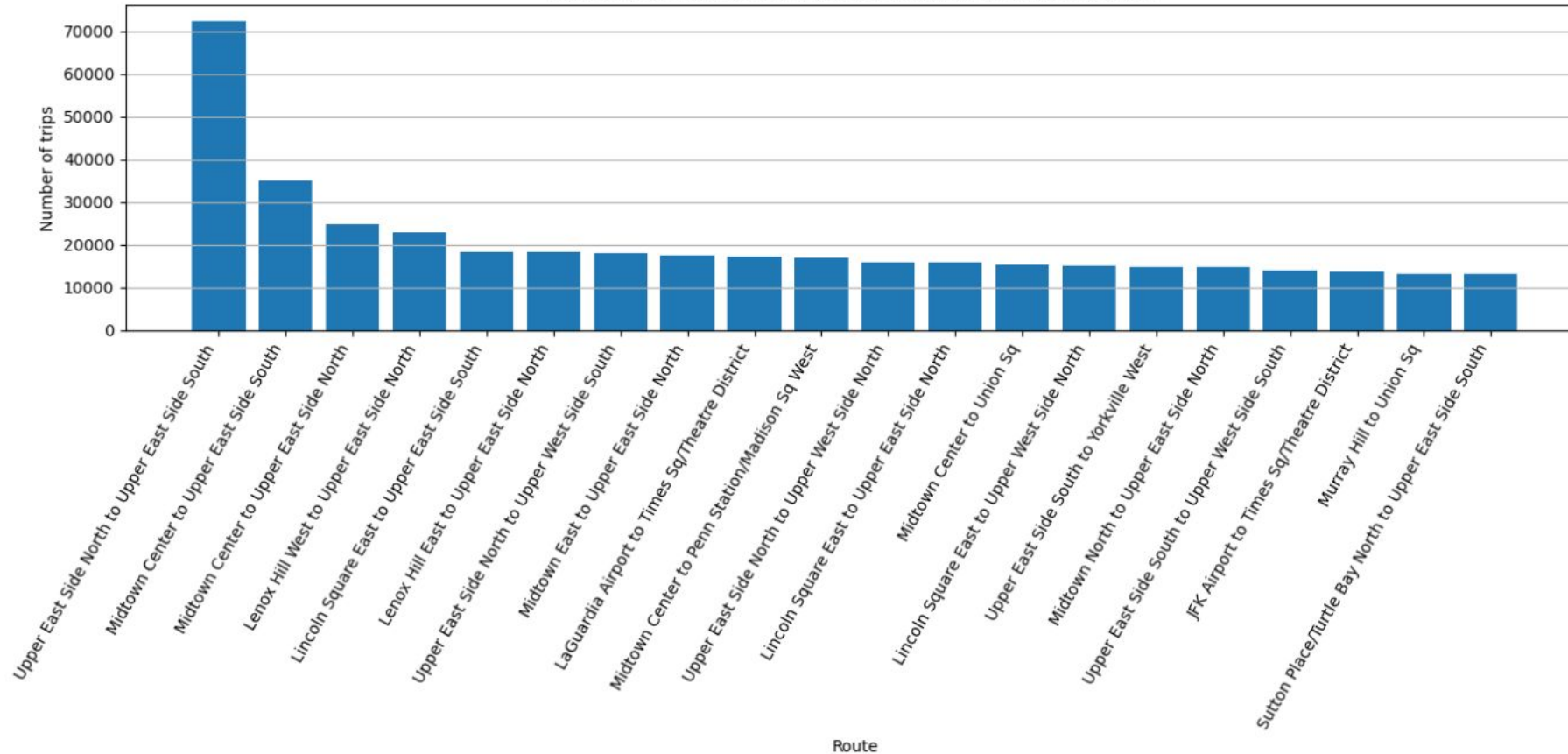
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Project Part 2: Bus Route Optimisation

RESULTS:

Figure 2: Top 20 routes by number of trips, 2025 (8am to 6pm)



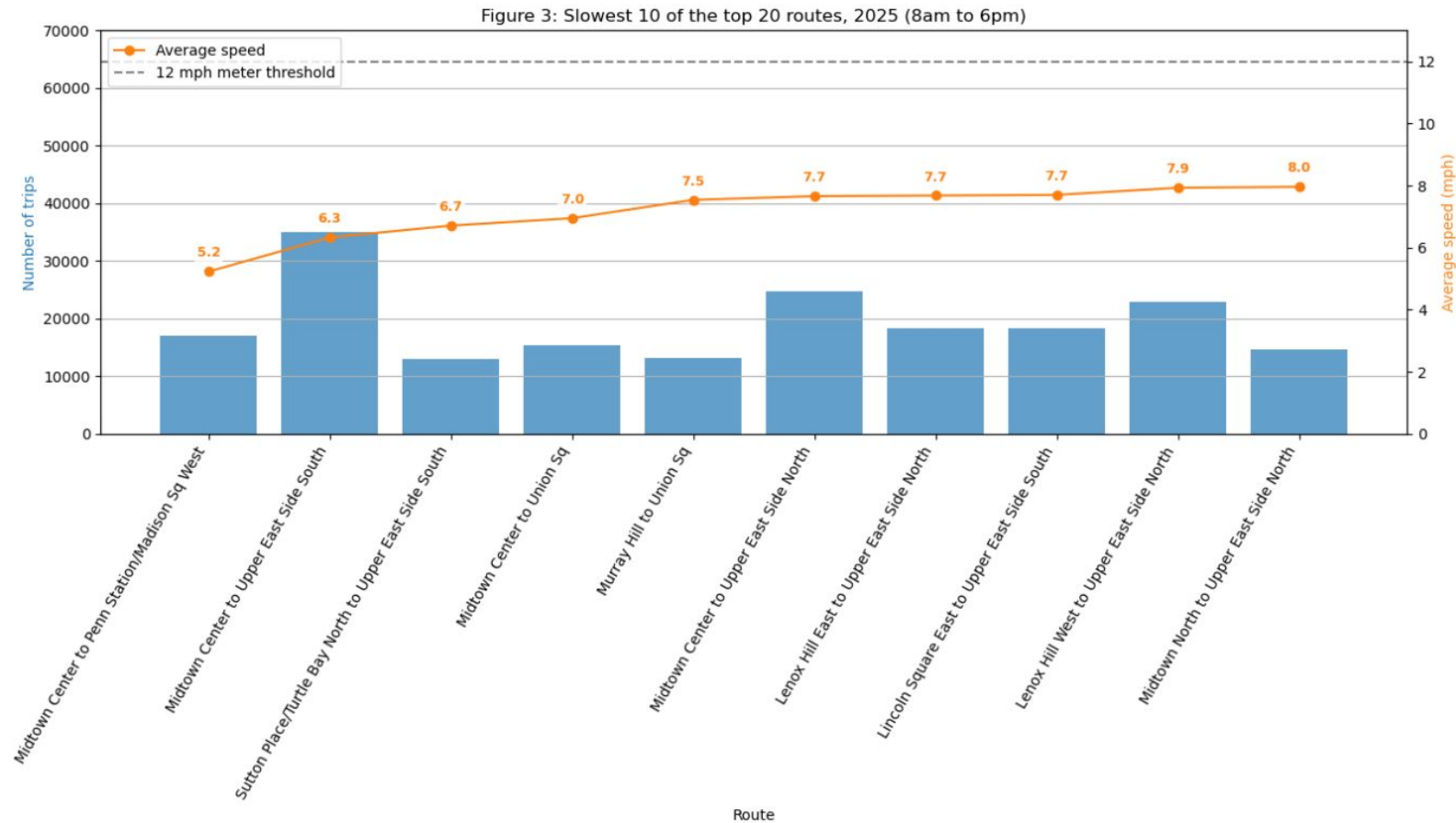
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RESULTS:



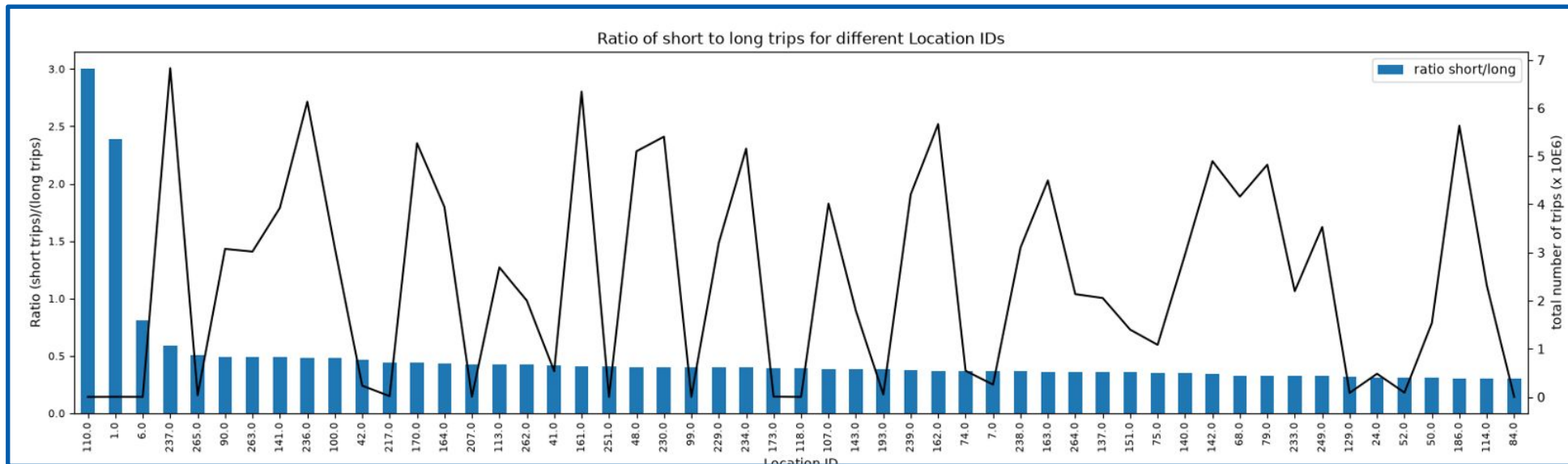
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Project Part 2: Bus Route Optimisation

RESULTS:



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Project Part 2: Bus Route Optimisation

RESULTS - Prioritised areas

FROM	TO
Midtown Center	Penn Station/Madison Sq West
Midtown Center	Upper East Side South
Sutton Place/Turtle Bay North	Upper East Side South
Midtown Center	Union Sq
Murray Hill	Union Sq
Midtown Center	Upper East Side North
Lenox Hill East	Upper East Side North
Lincoln Square East	Upper East Side North
Lenox Hill West	Upper East Side North
Midtown North	Upper East Side North



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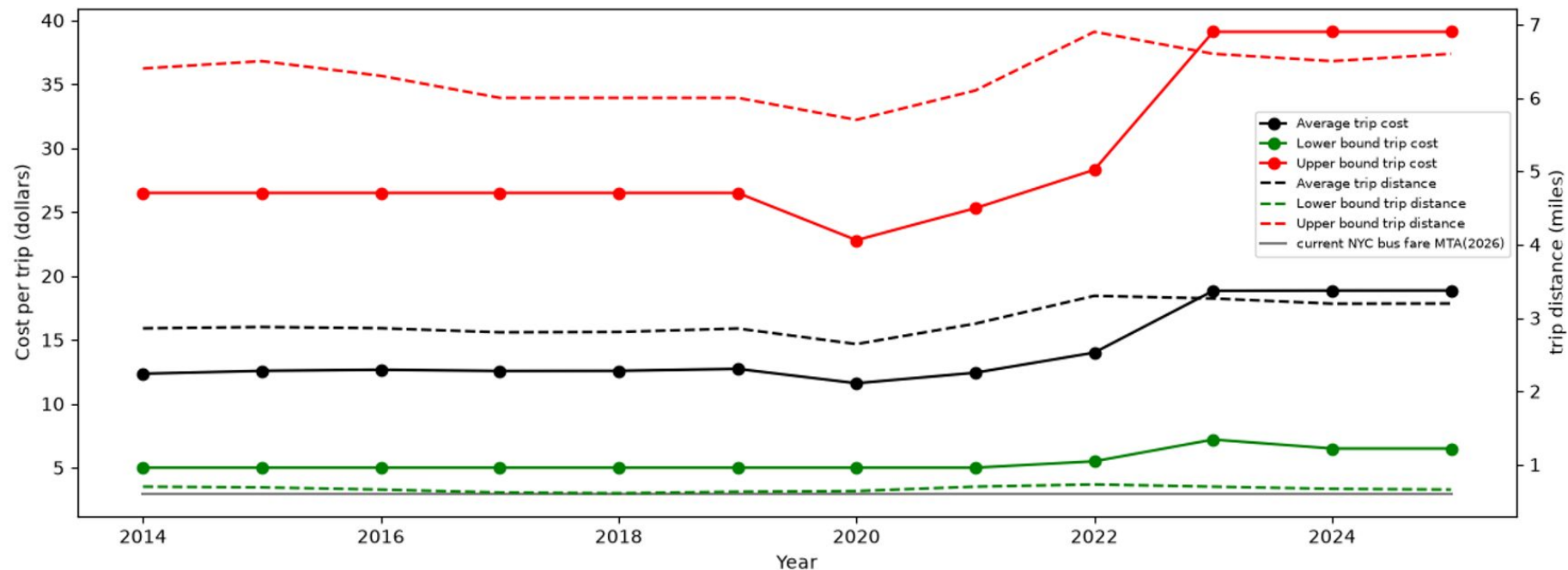
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Project Part 2: Bus Route Optimisation

RESULTS:

- Trends in terms of the costing



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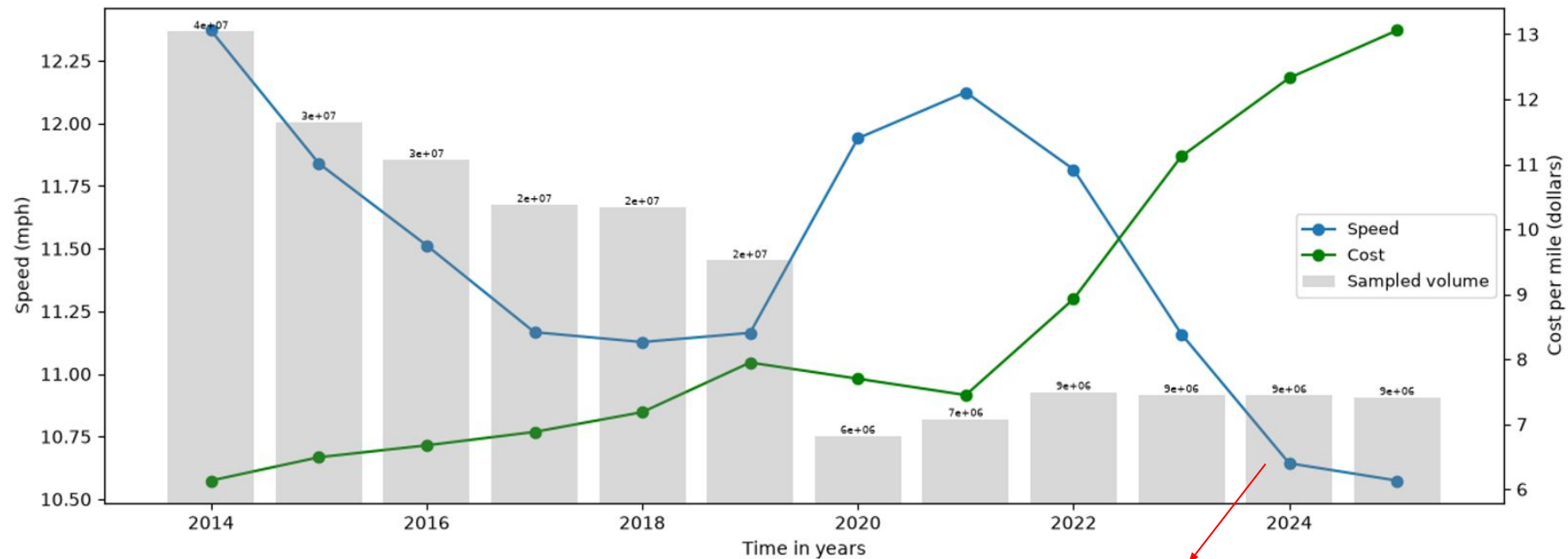
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Project Part 2: Bus Route Optimisation

RESULTS:

- Trends over time
- Annual trends



Slower speed over time =
increased congestion



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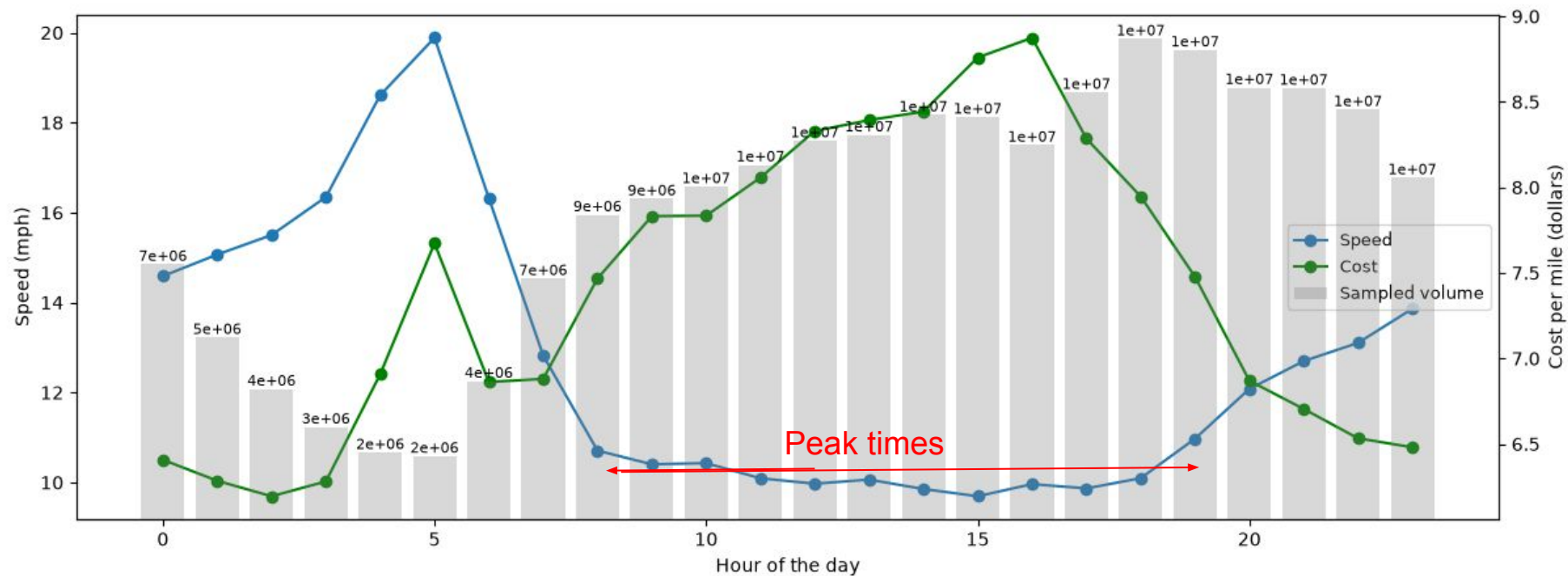
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Project Part 2: Bus Route Optimisation

RESULTS:

- hourly trends



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PySpark Implementation



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Project Part 2: Bus Route Optimisation

RESULTS:

- Hourly trends - PySpark versus traditional MapReduce

```
hourly_congestion_df = df.select(["hour_pickup", "average_trip_speed"])

hourly_speeds = (
    hourly_congestion_df
    .groupBy("hour_pickup")
    .agg(avg("average_trip_speed").alias("average_speed"))
    .orderBy("hour_pickup")
)

hourly_speeds.show()
```

```
hourly_cost_df = df.select(["hour_pickup", "trip_distance", "fare_amount"])
hourly_cost_df = hourly_cost_df.withColumn("cost_per_mile_h", col("fare_amount")/col("trip_distance"))
```

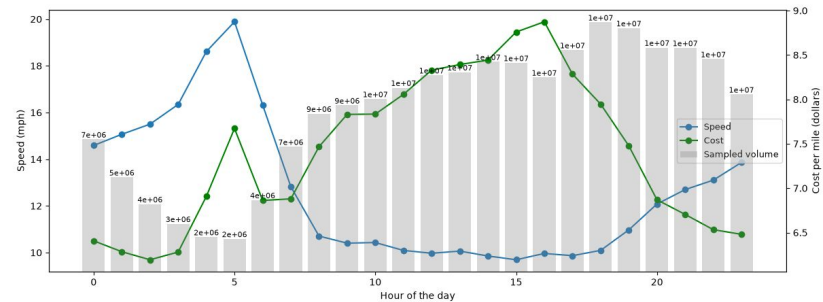
```
hourly_cost = (
    hourly_cost_df
    .groupBy("hour_pickup")
    .agg(avg("cost_per_mile_h").alias("cost_per_mile_h"))
    .orderBy("hour_pickup")
)

hourly_cost.show()
```

```
trip_vols_h_df = df.select(["hour_pickup", "passenger_count"])

trip_vols_hourly = (
    trip_vols_h_df
    .groupBy("hour_pickup")
    .agg(sum("passenger_count").alias("passenger_count"))
    .orderBy("hour_pickup")
)

trip_vols_hourly.show()
```



PySpark's `.groupBy` performs the function of mapping and shuffling the data according to the column identified as containing the key (i.e., "hour_pickup" in this case).

PySpark's `.agg` function and functions like `sum` or `average (avg)` perform the traditional MapReduce function of the Combiner and the Reducer:

- Local combining may occur in the partitions specified of sums and counts
- These are then brought together for the final Reduction
- The values of the k:v pairs are specified as the "average_trip_speeds", "cost_per_mile_h" and "passenger_count".
- These are lazy functions

PySpark's `.show()` prompts the action



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Project Part 2: Bus Route Optimisation

RESULTS:

- Execution plans for calculations shown on the previous slide:
- Hourly average speeds:

```
== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- Sort [hour_pickup#73 ASC NULLS FIRST], true, 0
   +- Exchange rangepartitioning(hour_pickup#73 ASC NULLS FIRST, 200), ENSURE_REQUIREMENTS, [plan_id=461]
      +- HashAggregate(keys=[hour_pickup#73], functions=[avg(average_trip_speed#13)], output=[hour_pickup#73, average_speed#679])
         +- Exchange hashpartitioning(hour_pickup#73, 200), ENSURE_REQUIREMENTS, [plan_id=458]
            +- HashAggregate(keys=[hour_pickup#73], functions=[partial_avg(average_trip_speed#13)], output=[hour_pickup#73, sum#685, count#686L])
               +- Project [hour(tpep_pickup_datetime#1, Some(Africa/Johannesburg)) AS hour_pickup#73, average_trip_speed#13]
                  +- FileScan parquet [tpep_pickup_datetime#1,average_trip_speed#13] Batched: false, DataFilters: [], Format: Parquet, Location: InMemoryFileIndex(2 paths)[file:/C:/Users/savan/Documents/MIT STUDIES/5. MIT 805/GROUP PROJECT/PA..., PartitionFilters: [], PushedFilters: [], ReadSchema: struct<tpep_pickup_datetime:timestamp_ntz,average_trip_speed:double>
```

Number of partitions

- Hourly trip volumes:

```
== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- Sort [hour_pickup#73 ASC NULLS FIRST], true, 0
   +- Exchange rangepartitioning(hour_pickup#73 ASC NULLS FIRST, 200), ENSURE_REQUIREMENTS, [plan_id=611]
      +- HashAggregate(keys=[hour_pickup#73], functions=[sum(passenger_count#3)], output=[hour_pickup#73, passenger_count#885])
         +- Exchange hashpartitioning(hour_pickup#73, 200), ENSURE_REQUIREMENTS, [plan_id=608]
            +- HashAggregate(keys=[hour_pickup#73], functions=[partial_sum(passenger_count#3)], output=[hour_pickup#73, sum#890])
               +- Project [hour(tpep_pickup_datetime#1, Some(Africa/Johannesburg)) AS hour_pickup#73, passenger_count#3]
                  +- FileScan parquet [tpep_pickup_datetime#1,passenger_count#3] Batched: false, DataFilters: [], Format: Parquet, Location: InMemoryFileIndex(2 paths)[file:/C:/Users/savan/Documents/MIT STUDIES/5. MIT 805/GROUP PROJECT/PA..., PartitionFilters: [], PushedFilters: [], ReadSchema: struct<tpep_pickup_datetime:timestamp_ntz,passenger_count:double>
```



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```
pair_counts = spark.sql("""
    SELECT p.zone_a,
           p.zone_b,
           za.Zone AS zone_a_name,
           zb.Zone AS zone_b_name,
           p.avg_speed,
           p.avg_distance,
           p.trips
    FROM (
        SELECT CASE WHEN PULocationID < DOLocationID THEN PULocationID ELSE DOLocationID END AS zone_a,
               CASE WHEN PULocationID < DOLocationID THEN DOLocationID ELSE PULocationID END AS zone_b,
               ROUND(AVG(average_trip_speed), 2) AS avg_speed,
               ROUND(AVG(trip_distance), 2) AS avg_distance,
               COUNT(*) AS trips
        FROM taxi_data
        WHERE PULocationID <> DOLocationID
              AND YEAR(tpcp_pickup_datetime) = 2025
              AND HOUR(tpcp_pickup_datetime) BETWEEN 8 AND 17
        GROUP BY 1, 2
        HAVING AVG(trip_distance) > 1
    ) p
    JOIN zones za ON p.zone_a = za.LocationID
    JOIN zones zb ON p.zone_b = zb.LocationID
    ORDER BY p.trips DESC
""")
```

CASE statements acts as the **MAPPER** by looking at each trip and turning it into a route key such that A to B and B to A become the same route

The GROUP BY clause acts as the **SHUFFLE** step by bringing all trips with the same zone pair together

COUNT() and AVG() act as our reduce functions by giving each route its volume of trips and average speed



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Conclusions



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CONCLUSIONS:

Recommendations to the NYC planning office:

Factor	Recommendation
Areas to Prioritise for rollout of bus routes and lanes	The areas listed in the table on the previous slide, in the order of occurrence in the table should be prioritised, especially those indicated in bold text, as in these areas congestion is worse, and many short trips are taken, that would be feasible as bus trips.
Scheduling of bus routes	More frequent scheduling is recommended during the hours identified as peak hours (8am to 18h00, especially at 15h00, where the highest congestion is reached for the day).
Ticket Prices	Required to remain below taxi trip prices for similar trip lengths, as was indicated in the chart
Payment Systems	Cards or scanning apps on which money can be loaded is recommended as the EDA completed in Part 1 indicated growth in the card payments and a decline in cash payments.



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